Latency-Failure-Aware Multi-Agent Fuzzy Reinforcement Learning for Reliable Service Function Chain Backup

Qian Zhou, *Member, IEEE,* Jiayang Wu, Fu Xiao, *Senior Member, IEEE,* and Yanchun Zhang, *Member, IEEE*

***Abstract*—Multi-access Edge Computing (MEC) environments are inherently dynamic and uncertain due to frequent user mobility and unpredictable edge node failures, which severely affect the reliability of Service Function Chains (SFCs). To address these challenges, a fuzzy-logic-based framework is proposed, integrating fuzzy objective transformation with a latency-failure-aware multi-agent fuzzy reinforcement learning (LFA-MAFRL) strategy. Our method utilizes ε-constrained fuzzy optimization to balance service acceptance and latency, and employs fuzzy inference mechanisms to enable robust communication and coordination among agents under uncertainty, mitigating the impact of noisy or partial information. Extensive experiments on the SFCSim simulation platform show that our approach achieves up to 9% improvement in key performance metrics such as availability and latency, compared to existing state-of-the-art methods.**

***Index Terms*—Fuzzy Mathematical Programming, Multi-Agent Reinforcement Learning, Service Function Chains, Multi-Access Edge Computing.**

I. Introduction

With the rapid development of the Internet of Things (IoT) and mobile communication technologies, Multi-Access Edge Computing (MEC) [1] has emerged by sinking computing, storage, and communication resources to the network edge, providing low-latency and highly reliable service guarantees for emerging applications such as unmanned aerial vehicles and vehicular networks [2]. In this context, Service Function Chain (SFC), as a core technology of Network Function Virtualization (NFV) [3], enables customized and efficient network services through the ordered connection of Virtual Network Functions (VNFs) [4], [5]. VNFs include firewalls, load balancers, intrusion detection systems, etc., which are decoupled through NFV technology, making network services more flexible and scalable [6], [7].

This work was supported by the National Natural Science Foundation of China under the grant Nos.62372244 and 61902199, and the Major Scientific Instruments and Equipments Development Project of National Natural Science Foundation of China under grant (62427809), National Key Research and Development Program of China under Grant (2024YFB2906700), Key Program of Natural Science Foundation of Jiangsu under grant (BK20243053). *(Corresponding author: Fu Xiao.)*

Qian Zhou, Fu Xiao are with the School of Computer Science, Nanjing University of Posts and Telecommunications, Nanjing 210003, China (e-mail: zhouqian@njupt.edu.cn; xiaof@njupt.edu.cn).

Jiayang Wu is with the School of Modern Posts, Nanjing University of Posts and Telecommunications, Nanjing 210003, China (e-mail: 1223096926@njupt.edu.cn).

Yanchun Zhang is with Zhejiang Normal University, Jinhua 321004, and the Institute for Sustainable Industries and Liveable Cities, Victoria University, Melbourne, VIC 3011, Australia (e-mail: yanchun.zhang@vu.edu.au).

However, the dynamic nature and resource constraints of edge networks pose significant challenges to the continuous service of SFCs [8]. User mobility leads to frequent handovers of service access points [9], [10], and failures of edge nodes may cause service interruptions [11], all of which reduce the availability of SFCs in certain areas [12]–[14]. Traditional static or single-objective resource allocation strategies are inadequate for meeting Quality of Service (QoS) requirements under these dynamic and uncertain conditions [15]. Therefore, it is crucial to study delay- and failure-aware SFC backup [16], [17], aiming to optimize SFC backup strategies such that user service latency is minimized while overall availability is maximized, even in the presence of mobility and node failures [18], [19].

Reinforcement Learning (RL), based on dynamic interactions between agents and environments, achieves sequential decision-making optimization by maximizing long-term cumulative rewards. It can adaptively learn strategies through trial and error in dynamic and uncertain environments. Existing RL approaches can be divided into model-free methods (e.g., Deep Q-Networks), which directly learn value functions or policy functions to handle large-scale state spaces; value function and policy gradient methods (e.g., Q-Learning [20], [21]), which are limited by their capacity to express discrete spaces and prone to convergence to local optima; and hybrid methods (e.g., Actor-Critic [22]), which combine value function approximation but rely on accurate state representation and exhibit limited generalization capability. These methods generally lack sufficient generalization in large-scale scenarios, and nonlinear function approximators struggle to ensure convergence to globally optimal policies, especially when the state space dynamically expands, often leading to local optima [23]. Some studies have extended RL to multi-agent systems, but each agent operates in a relatively independent, partially observable environment [24]. Even if inter-agent communication is allowed, the received information may contain observation noise or policy uncertainty from other agents, degrading decision quality [25].

Fuzzy mathematics addresses uncertainty and imprecision, with its core comprising fuzzy mathematical programming and fuzzy inference systems. Fuzzy mathematical programming can handle problems where the objective function or constraints are fuzzy and uncertain by defining membership functions for fuzzy constraints to quantify the degree of satisfaction. Fuzzy inference systems are intelligent decision-making models based on fuzzy logic, involving fuzzification of inputs, application of fuzzy rule bases, aggregation of rule outcomes, and defuzzification to output precise control quantities [26]. Existing studies have combined traditional fuzzy logic reasoning with RL; however, these methods require manual design of membership functions and fuzzy rules based on human prior knowledge, which reduces overall model performance and limits autonomous exploration in dynamic and uncertain environments [27]. To address the partial observability and local uncertainty in SFC environments, fuzzy mathematics is integrated into multi-agent reinforcement learning to enhance agents' decision-making, mitigate communication noise, and enable robust local learning fusion, which is called Multi-Agent Fuzzy Reinforcement Learning (MAFRL) [28].

Therefore, current SFC backup methods still fall short in adaptability to dynamic environments and multi-objective optimization [29], [30]. Motivated by the increasing number of mobile users and the inherent unreliability of edge nodes, this paper introduces a multi-agent fuzzy reinforcement learning framework, incorporating fuzzy inference-based communication and ε-constraint multi-objective optimization to address delay- and failure-aware SFC backup. The key contributions of this work are summarized as follows:

1) A target transformation method based on fuzzy mathematical programming is proposed. The ε-constraint method is combined with fuzzy membership functions to achieve dynamic multi-objective optimization between user service hit rewards and the overall network latency.
2) A Latency-Failure-Aware Multi-Agent Fuzzy Reinforcement Learning (LFA-MAFRL) method is proposed, with a fuzzy inference-based inter-agent communication mechanism designed to reduce observation noise and policy uncertainty from other agents in uncertain environments. This facilitates efficient collaborative learning of user mobility and node failures, ensuring SFC availability within specific geographical regions. Theoretical guarantees are also provided.
3) Experimental evaluations on the open-source simulation platform SFCSim demonstrate performance improvements across multiple metrics. The results validate that the proposed method effectively adapts to dynamic and uncertain network environments, enhances the efficiency of SFC backup, and ensures high system availability and low latency.

II. Related Work

A. Deployment and Backup of Service Function Chains

In the deployment and backup of Service Function Chains (SFCs), numerous studies have proposed various approaches to ensure high availability, optimize resource utilization, and enhance adaptability in dynamic environments. Traditional research primarily focuses on static resource allocation in data center environments. However, with the rise of Multi-Access Edge Computing (MEC) in recent years, research attention has gradually shifted toward MEC environments characterized by resource constraints and dynamically changing demands.

Shang et al. [31] proposed an adaptive deployment scheme that combines static and dynamic backup strategies to improve the efficiency of SFC backup in both edge and cloud computing environments. This scheme enhances adaptability by optimizing resource utilization and adjusting backup strategies based on the failure rate of Virtual Network Functions (VNFs). However, it relies on a preset failure rate adjustment mechanism, making it insufficiently responsive to unknown environmental changes, and its performance is limited under extreme resource constraints. Wang et al. [11] introduced an online learning-based SFC backup optimization method that adjusts backup strategies to maximize backup hit rewards. Although the method incorporates online learning, it only focuses on changes in VNF demand, without considering the dynamic variation of failure rates. Moreover, it exhibits slow convergence and lacks responsiveness to sudden failures. Madani et al. [32] proposed an SFC backup method based on adaptive multipath routing, which allocates traffic across multiple paths to improve fault tolerance and reduce resource consumption. While this method achieved approximately 20% savings in bandwidth and computing resources, it still struggles to adapt to the rapid changes in mobile edge computing environments. Wu et al. [37] proposed an adaptive SFC orchestration scheme for UAV-enabled low-altitude intelligent networks based on a dueling double deep Q-network with regularization (D3QNReg). By integrating the fuzzy analytic hierarchy process to model QoS–QoE relationships and formulating the orchestration process as a Markov decision process, the method enhances training stability and improves energy efficiency and task completion. However, it mainly focuses on single-agent optimization and does not explicitly address multi-agent cooperation or multi-objective trade-offs among latency, service availability, and reliability, limiting its applicability in complex dynamic environments.

In summary, existing SFC deployment and backup methods have made some progress in ensuring high availability and resource optimization. However, most of these approaches fail to adequately consider the dynamic changes in service function chain demand and failure rates within a given region, making it difficult to maintain efficient backup strategies in real MEC environments. Furthermore, many methods rely on static optimization mechanisms and cannot quickly adapt to environmental changes.

B. Fuzzy Reinforcement Learning

With the increasing prevalence of intelligent systems, fuzzy reinforcement learning (FRL) has been widely applied across various fields. FRL, in particular, demonstrates excellent adaptability and control capabilities in dynamic environments. By handling continuous and complex state spaces, FRL significantly enhances the practical applicability of traditional reinforcement learning.

Salari et al. [33] proposed a multi-agent system based on FRL for real-time energy management in residential microgrids. Through agent collaboration and shared state variables, the system optimizes electric vehicle (EV) charging and energy storage, ensuring power supply while reducing electricity costs. Experimental results showed that the method improved the health of the storage system and reduced electricity prices.

This article has been accepted for publication in IEEE Transactions on Mobile Computing. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TMC.2026.3688952

TABLE I: Summary of Related Work on SFC Deployment/Backup and Fuzzy Reinforcement Learning

Reference	Key Ideas	Limitations	Area
[31]	Static-dynamic backup across edge-cloud; failure-aware strategy	Depends on fixed failure model; poor adaptability	SFC
[11]	Online learning-based backup under demand variation	Slow convergence; ignores failure dynamics	SFC
[32]	Multipath routing to enhance fault tolerance	Limited MEC adaptability	SFC
[33]	Multi-agent FRL for EV/energy management	Poor efficiency-supply trade-off; no cooperation modeling	FRL
[34]	FRL for 6G-V2X resource allocation	Ignores latency; lacks multi-goal trade-off	FRL
[35]	Online FRL for dynamic energy strategy in vehicles	Weak energy-stability balance; no agent interaction	FRL
[36]	FRL-based VM scheduling for energy saving	No SLA/efficiency trade-off modeling	FRL
[37]	D3QN-based SFC orchestration for QoE-aware UAV networks	No multi-agent coordination or multi-objective trade-off	FRL

However, it failed to effectively balance multiple objectives such as energy supply and EV charging efficiency, and did not fully consider collaboration and competition among agents. Zhang et al. [34] proposed a dynamic resource allocation model that integrates fuzzy logic and reinforcement learning for vehicle-to-everything (V2X) communication in 6G networks. The model optimized network throughput and reduced interference, outperforming traditional algorithms in convergence speed and throughput. However, it did not address multi-objective trade-offs in a multi-agent environment and failed to meet QoS requirements such as low latency and high reliability. Yang et al. [35] developed an FRL-based online energy management strategy for energy distribution in fuel cell vehicles. By incorporating road condition recognition technology, this method dynamically adjusted the energy management strategy to reduce fuel consumption and system degradation. Nevertheless, it did not adequately address collaboration and conflict among multiple agents, nor did it balance energy efficiency and system stability. Yu et al. [36] proposed an FRL-based virtual machine (VM) scheduling strategy for cloud resource management. By combining fuzzy control with reinforcement learning, this approach optimized VM placement and migration, reducing energy consumption and improving resource utilization. However, it failed to consider multi-objective optimization, and did not simultaneously optimize energy consumption, service-level agreements, and resource utilization, thus falling short of practical requirements.

Although FRL has achieved promising results in various application domains, most existing research primarily focuses on optimizing single objectives without fully addressing the need to balance multiple, often conflicting, goals. As a result, many methods struggle to find optimal solutions that consider the trade-offs between different objectives, such as service availability, network latency, and resource consumption. Moreover, a significant number of approaches fail to effectively model the cooperation and competition dynamics among multiple agents, which can hinder performance, particularly in complex and dynamic environments. This limitation often leads to poor adaptability and robustness when the system faces real-world uncertainties, such as changing network conditions, varying workloads, and unexpected failures [38]. As highlighted in Table I, these challenges remain critical obstacles in the development of more effective and scalable FRL-based solutions.

III. Proposed Work

A. Overall Framework

In a multi-access edge computing (MEC) environment, the deployment and backup of Service Function Chains (SFCs) must address the challenges posed by user mobility and edge node failures. As the user's location changes, the optimal service access point may migrate accordingly. Meanwhile, failures of computing nodes may lead to the unavailability of certain Virtual Network Functions (VNFs), thereby disrupting the operation of the entire SFC. Therefore, in such a dynamic network environment, it is essential to predict potential user migration paths in advance and to complete the backup and migration of SFCs before critical node failures occur, ensuring continuous and high-quality service delivery.

Figure 1 illustrates a typical scenario in MEC, where a user initially accesses the network via a drone and requests an SFC composed of VNF1, VNF2, and VNF3. At this stage, the user is located in region D_1, and the nearest edge node is n_2. Thus, the instantiation of the SFC is deployed such that n_1 hosts VNF1 and VNF2, while n_2 hosts VNF3. As the user moves towards region D_2, failure to implement targeted optimization may result in an inability to meet the complete service demand within D_2, especially when node n_4 fails at time $t = 3$, leading to insufficient computational resources for hosting VNF3. Without prior backup of VNF3, the user would be unable to access the full SFC upon arriving at D_2, thereby degrading the quality of network service. However, if the system detects user mobility trends between $t = 2$ and $t = 3$, and selects suitable nodes for backup migration based on an optimization strategy, it is possible to complete the deployment of VNF3 before the failure of n_4. Consequently, at $t = 4$, region D_2 can still provide the complete SFC. As a result, when the user accesses node n_6 via a car, they can still enjoy a Quality of Service (QoS) experience similar to that in region D_1.

This study employs a fuzzy mathematical programming approach to transform a multi-objective optimization problem into a single-objective one that is easier to solve. The backup and migration of SFCs involve two conflicting objectives:

This article has been accepted for publication in IEEE Transactions on Mobile Computing. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TMC.2026.3688952

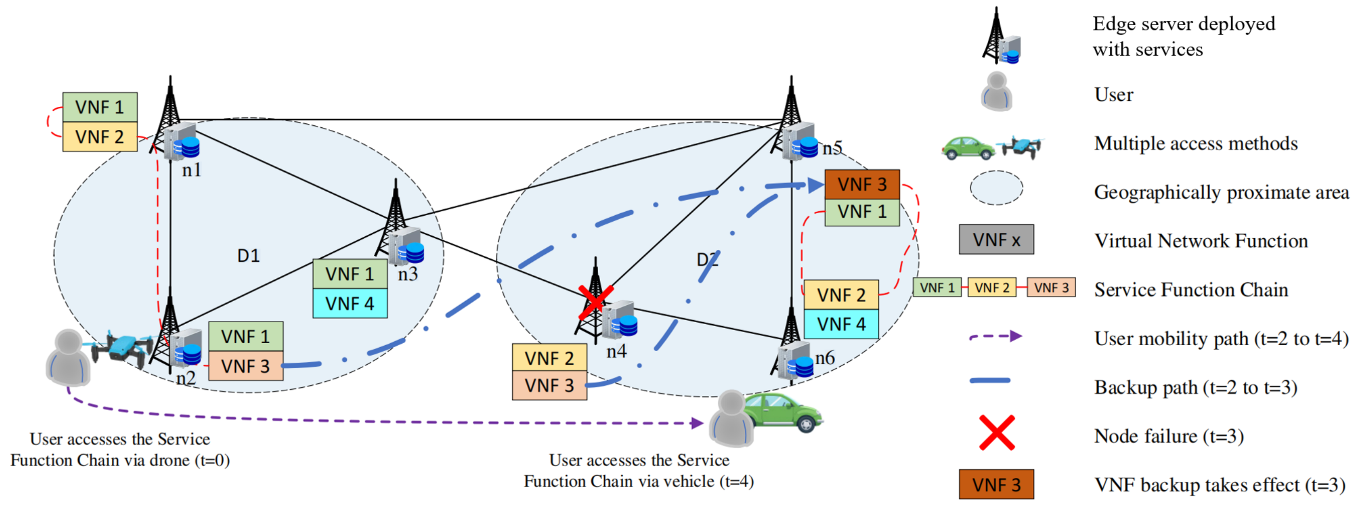


Fig. 1: Overall Framework.

QoS assurance and resource consumption. A naive weighted-sum method may result in solutions biased toward extreme points, thereby reducing system adaptability. To address this, an ε-constraint method is adopted to balance the QoS and resource objectives, enabling agents to maximize the reward for successful service delivery while constraining the overhead induced by migration in terms of computing and storage resources. On this basis, a fuzzy inference system is integrated to enable multi-agent collaboration and optimize backup migration decisions.

During the backup migration process, each agent first receives an input model and transforms its objectives through fuzzy membership functions, where the optimal and nadir values are calculated. Membership functions are employed to convert local observations into fuzzy variables, while constraints are imposed to ensure feasible decision spaces. Within the fuzzy reinforcement learning training phase, local observations are fuzzified, and fuzzy rules are applied to infer migration actions. To cope with communication noise, fuzzy inference integrates both local and shared states, thereby enhancing the generalization ability of decision-making across agents. The defuzzification step transforms fuzzy inference outputs into concrete migration actions. Neural components are incorporated into the agent architecture to further improve approximation capability and policy optimization. Finally, during dynamic migration, the system identifies failure nodes, pending backup nodes, and storage nodes, while constructing backup paths to minimize service disruption. By integrating fuzzy reasoning with reinforcement learning, the agents can dynamically adjust backup locations and achieve efficient cooperative migration in the edge network, as illustrated in Figure 2.

The primary notations used throughout this paper are summarized as Table II.

B. Problem Modeling

In edge computing environments, the backup deployment problem of Service Function Chains (SFCs) can be formulated as an optimization problem that aims to maximize user service hit rate while minimizing link delay under resource constraints. Following the scenarios in [36], the edge environment is modeled as an undirected connected graph $G = (N, E)$, where N represents the set of edge servers and E denotes the set of network links. To reduce delay in SFC backup deployment, frequently accessed SFCs are considered for placement in low-latency locations according to their popularity.

Definition 1 (Regional SFC Popularity). Let binary variable $Q_{k,f}(t)$ indicate whether user k requests SFC f in time slot t. The total requests for SFC f is:

$$Q_f(t) = \sum_{k \in K} Q_{k,f}(t) \tag{1}$$

with mathematical expectation (popularity):

$$q_f(t) = \mathbb{E}\left[Q_f(t)\right] \tag{2}$$

Definition 2 (SFC Failure Rate). Let $V_i(t)$ represent the failure rate of virtual network function i in time slot t, with expectation:

$$v_i(t) = \mathbb{E}\left[V_i(t)\right] \tag{3}$$

For each SFC f, let I_f denote the set of VNFs that constitute the service function chain f. The overall failure rate of SFC f is then:

$$U_f(t) = 1 - \prod_{i \in I_f} (1 - v_i(t)) \tag{4}$$

Definition 3 (Link Delay). Let $l(u, v)$ denote the link delay between servers u and v. The total delay for SFC f is:

$$L_f(t) = \sum_{j=1}^{|I_f|-1} \sum_{(u,v) \in E} \phi^j_{f(u,v)} l(u, v) \tag{5}$$

Definition 4 (Service Hit Reward). The reward obtained when users access available SFCs from their nearest edge node within geographic proximity.

This article has been accepted for publication in IEEE Transactions on Mobile Computing. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TMC.2026.3688952

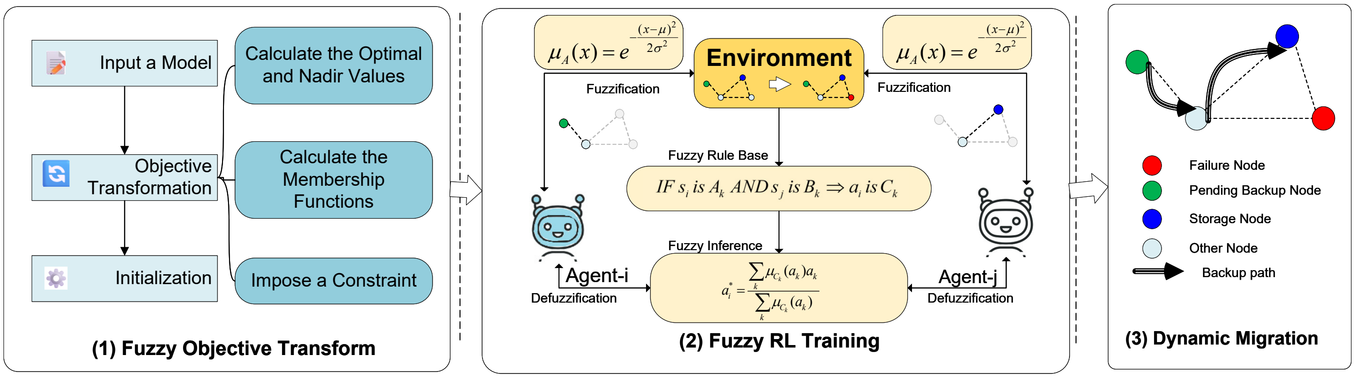


Fig. 2: Work Flow of Latency-Failure-Aware Multi-Agent Fuzzy Reinforcement Learning (LFA-MAFRL).

Our optimization objectives under resource constraints are:

$$\max \frac{1}{T}\sum_{t=0}^{T-1}\sum_{f\in F}\mathbb{E}\left[(\omega Q_f(t)-\mu L_f(t))\cdot X_f(t)\cdot(1-U_f(t))\right] \tag{6}$$

$$\min L_f(t) \tag{7}$$

where ω, μ are constants balancing popularity and delay impacts. The backup integrity requirement is:

$$X_f(t) = 1-\min\left(1, \sum_{i\in I_f}(1-P_{i,n}(t))\right) \tag{8}$$

with resource constraints:

$$\sum_{i\in J} D_i\cdot P_{i,n}(t)\le M_n, \quad \forall n\in N \tag{9}$$

This NP-hard problem requires fuzzy reinforcement learning to estimate popularity and failure rates for deployment.

Definition 5 (Inter-Agent Communication). In multi-agent systems, this refers to agents sharing local observations and decision states to collaboratively optimize global objectives. Our mechanism employs fuzzy inference to convert service request intensity and backup status into fuzzy linguistic variables, reducing communication overhead and noise while enhancing generalization.

C. Objective Transformation Based on Fuzzy Mathematical Programming

Maximizing user service hit reward and minimizing total environmental latency are two mutually constrained optimization objectives. In multi-objective optimization problems, a common approach is to convert multiple objectives into a single objective for optimization to enhance computational efficiency and convergence speed. However, the simple weighted sum method tends to concentrate the solution space on extreme solutions, i.e., the corner points of the Pareto front, thereby reducing solution diversity. Furthermore, under the reinforcement learning framework, fixed weight selection may fail to adapt to dynamically changing environments, necessitating more flexible objective transformation strategies.

It is important to emphasize that the service function chain backup deployment problem considered in this work is inherently non-convex. First, the deployment decision variables are discrete (binary backup placement indicators), leading to a combinatorial feasible set $\Omega\subset\{0,1\}^K$, which is non-convex by definition. Second, the reward function $R(y)$ and latency function $L(y)$ involve nonlinear coupling among resource allocation, link congestion, and failure probabilities. For example, congestion-induced latency exhibits nonlinear growth under resource saturation, and the reward formulation includes multiplicative interaction terms between request intensity, backup state, and failure rate. Therefore, both the feasible region and the objective mappings contribute to a potentially non-convex Pareto front.

Under such conditions, the weighted sum method can only recover Pareto optimal solutions located on the convex hull of the objective space, while failing to capture non-supported Pareto points in concave regions. In contrast, the ε-constraint method does not rely on convexity assumptions and can generate Pareto optimal solutions even when the Pareto front is non-convex.

To avoid the emergence of extreme solutions, the ε-constraint method is adopted, which selects a primary optimization objective and transforms the other objective into a constraint. By adjusting the ε values, the feasible solution range is modified. This method effectively alters the shape of the feasible region, thereby generating non-corner Pareto optimal solutions and enhancing solution diversity. Additionally, each optimization iteration may yield distinct effective solutions, leading to a richer solution set.

Under the fuzzy mathematical programming framework, fuzzy membership functions are utilized to describe the satisfaction levels of different objectives. Through these membership functions, diverse physical quantities can be normalized, enabling unified optimization of objective functions across varying scales. In the problem formulation, we construct the membership function for user service hit reward $m_R(y)$ and the membership function for total latency $m_L(y)$, both ranging within $[0,1]$. A value of 1 indicates complete satisfaction of the objective, while 0 signifies total dissatisfaction. Initially, the single-objective optimal values are determined independently to construct reference extreme points in the objective space. Specifically, the reward-optimal value is defined as

$$R^{(\mathrm{opt})} = \max_{y\in\Omega} R(y), \tag{10}$$

TABLE II: Primary Notations

Symbol	Definition
$G=(N,E)$	Edge network graph
N	Set of edge servers (nodes)
E	Set of network links
$Q_{k,f}(t)$	Request indicator of user k for SFC f at time t
$Q_f(t)$	Total requests of SFC f at time t
$q_f(t)$	Popularity of SFC f
$V_i(t)$	Failure rate of VNF i at time t
$v_i(t)$	Expected failure rate of VNF i
I_f	Set of VNFs composing SFC f
$U_f(t)$	Failure rate of SFC f
$l(u,v)$	Link delay between nodes u and v
$L_f(t)$	Total delay of SFC f
$\phi^j_{f(u,v)}$	Link usage indicator of the j-th hop
$X_f(t)$	Backup integrity indicator of SFC f
$P_{i,n}(t)$	Placement indicator of VNF i on node n
D_i	Resource demand of VNF i
M_n	Available resource capacity of node n
ω, μ	Reward balancing coefficients
$R(y)$	Service hit reward function
$L(y)$	Total latency function
$m_R(y)$	Membership function of reward
$m_L(y)$	Membership function of latency
$R^{(\text{opt})}$	Optimal reward value
$L^{(\text{opt})}$	Optimal latency value
$R^{(\text{nad})}$	Nadir reward value
$L^{(\text{nad})}$	Nadir latency value
$\tilde{G}_R, \tilde{G}_L$	Fuzzy goal sets
$o_n(t)$	Local observation of agent n
$O^F_n(t)$	Fuzzy state vector of agent n
$FIS(\cdot)$	Fuzzy inference system
$\mu_A(x)$	Membership function of fuzzy set A
μ, σ	Parameters of Gaussian membership function
s_i, s_j	Local state of agent i and neighbor j
$\tilde{s}_j$	Fuzzy state received from neighbor j
β	Local–neighbor state fusion coefficient
ω_{ij}	Communication weight from agent j to i
$\mathcal{N}_n$	Communication neighborhood of agent n
$C^F_n(t)$	Aggregated fuzzy communication message
$W_{n,m}$	Communication weight between agents
$S_n(t)$	Final input state of agent n
α	Local–global state balancing coefficient
$R_n(t)$	Immediate reward of agent n
$\tilde{R}_n(t)$	Fuzzy reward used for learning
$\mu_R(\cdot)$	Reward membership function
$\mu_Q(\cdot)$	Request intensity membership function
$\mu_L(\cdot)$	Latency membership function
β_q	Popularity update rate
β_u	Failure rate update rate
π_n	Policy of agent n
$J(\pi_n)$	Expected return of policy
θ	Policy network parameters
a_n	Action selected by agent n
$\mathcal{A}_n(t)$	Action space of agent n
$A_{\max}$	Maximum action dimension
$M_n(t)$	Binary action mask
z_i	Logit output of policy network

where Ω denotes the feasible deployment set under resource and topology constraints. Let $y^*_R \in \arg\max_{y\in\Omega} R(y)$ be the corresponding optimizer. The associated latency value is then defined as

$$L^{(\text{nad})} := L(y^*_R). \tag{11}$$

Similarly, for latency minimization,

$$L^{(\text{opt})} = \min_{y\in\Omega} L(y), \tag{12}$$

with optimizer y^*_L, and the corresponding reward value

$$R^{(\text{nad})} := R(y^*_L). \tag{13}$$

It is important to clarify that this relationship is not a universal functional dependency between objectives. Instead, these values serve as extreme reference points derived from single-objective optimization under fixed system constraints. They provide upper and lower performance bounds for normalization in the fuzzy programming framework.

In resource-limited MEC environments, maximizing service hit reward typically requires deploying additional backup instances or placing services closer to user clusters, which may increase resource competition and link congestion. Consequently, the induced latency value reflects a structural trade-off under constrained optimization, but it does not imply strict monotonicity or universal validity across arbitrary system configurations.

Define the fuzzy goal sets $\tilde{G}_R$ and $\tilde{G}_L$, whose membership functions $m_R(y)$ and $m_L(y)$ must satisfy the following axiomatic conditions:

- $m_R(y)$ monotonically increases as $R(y)$ increases, while $m_L(y)$ monotonically increases as $L(y)$ decreases;
- $m_R(y), m_L(y) \in [0,1]$, attaining 1 when the objective reaches its ideal value and 0 when it reaches the tolerable lower bound;
- The fuzzy feasible region is a convex set, guaranteeing the existence and uniqueness of the optimal solution.

Linear piecewise functions can be employed to construct membership functions with minimal complexity that satisfy the aforementioned conditions. Let $R^{(\text{opt})}$ and $L^{(\text{opt})}$ denote the single-objective optimal values, and $R^{(\text{nad})}, L^{(\text{nad})}$ represent the Nadir points (worst-case values). The normalization formulas are derived as:

$$m_R(y) = \frac{R(y) - R^{(\text{nad})}}{R^{(\text{opt})} - R^{(\text{nad})}}, \tag{14}$$

satisfying $m_R = 1$ when $R(y) = R^{(\text{opt})}$ and $m_R = 0$ when $R(y) = R^{(\text{nad})}$, with linear interpolation ensuring convexity. Similarly,

$$m_L(y) = \frac{L^{(\text{nad})} - L(y)}{L^{(\text{nad})} - L^{(\text{opt})}}, \tag{15}$$

satisfies $m_L = 1$ when $L(y) = L^{(\text{opt})}$ and $m_L = 0$ when $L(y) = L^{(\text{nad})}$, with negative derivatives guaranteeing monotonicity.

Theorem 1 (Pareto Effectiveness). Using the aforementioned membership functions, the solution generated via the ε-constraint method $\max m_R(y)$ subject to $m_L(y) \geq \varepsilon$ is guaranteed to be a weak Pareto optimal solution for the original multi-objective optimization problem [39].

Proof. If the constraint set $S(y)$ is convex and the objective functions satisfy quasi-convexity, the ε-constraint method can generate the Pareto front. Since linear membership functions are strictly monotonic, the Pareto front of the original multi-objective problem maintains a homeomorphic mapping with the solution space of the fuzzy optimization problem. Suppose there exists a solution y^* superior to the current solution y'; then it must hold that $m_R(y^*) > m_R(y')$ and $m_L(y^*) \geq m_L(y')$, contradicting the assumption that y' is the optimal solution of the constrained optimization problem.

To optimize both objectives during the reinforcement learning process, the ε-constraint method is applied by designating the hit reward objective function as the optimization target while constraining the membership degree of total latency to be no less than a threshold ε:

$$m_L(y) - \varepsilon \geq 0, \quad y \in S(y), \tag{16}$$

where $S(y)$ represents the constraint set of the original problem, and ε ranges within $[0,1]$, gradually increasing during the optimization process. In each optimization iteration, ε in the constraint acts as the intensity of the total latency requirement. A larger ε implies stricter demands on total latency, while a smaller ε permits higher service hit rewards. This ensures that the agent maintains total latency within acceptable limits while optimizing service hit rewards.

D. Multi-Agent Fuzzy Reinforcement Learning

In multi-agent reinforcement learning environments, each agent needs to make decisions in partially observable environments and optimize global rewards through communication and collaboration. In the service function chain backup optimization problem, agents must not only select optimal backup strategies based on local observations but also share information with other agents to reduce decision uncertainty. However, directly sharing numerical states may incur high communication overhead and introduce noise. Therefore, fuzzy inference systems are introduced to manage communication uncertainty and enhance system generalization through fuzzy reinforcement learning [40].

Consider a system composed of multiple edge servers, where each server n acts as an agent using fuzzy communication mechanisms to share locally observed service request intensities and backup states of service function chains. Each agent n observes local state $o_n(t)$ at timestep t and computes a fuzzy state vector:

$$O_n^F(t) = FIS\left(o_n(t)\right) \tag{17}$$

where $FIS(\cdot)$ represents the fuzzy inference system comprising fuzzification, fuzzy rules, and defuzzification. Let x be the input variable and $\mu_A(x)$ the membership function. The fuzzification process uses Gaussian functions:

$$\mu_A(x) = \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) \tag{18}$$

where μ is the center and σ the standard deviation. For fuzzy rules, let agent i observe local state s_i and neighbor states $s_j (j \in N_i)$ within communication neighborhood N_i. Fuzzy rule R_k is defined as:

R_k: IF s_i is A_k AND s_j is B_k, THEN a_i is C_k

Here, A_k, B_k, and C_k denote fuzzy sets defined on the domains of the local state s_i, the neighbor state s_j, and the output action a_i, respectively. The fuzzy sets A_k and B_k characterize linguistic terms associated with the input variables, while C_k represents the fuzzy conclusion of the rule.

The defuzzified output a_i^* is computed via the centroid method:

$$a_i^* = \frac{\sum_k \mu_{C_k}(a_k)\, a_k}{\sum_k \mu_{C_k}(a_k)} \tag{19}$$

Assuming communication messages contain high-frequency noise components, agent i observes local state s_i and receives fuzzy communication $\tilde{s}_j$ from neighbors $j \in N_i$:

$$\tilde{s}_j = \sum_k \mu_k s_j^k \tag{20}$$

where μ_k denotes fuzzy weights (computed by the fuzzy inference system) and s_j^k represents the value of agent j's state belonging to fuzzy set A_k. Agent i fuses states as:

$$s_i = \beta s_i + (1-\beta) \sum_{j \in N_i} \omega_{ij} \tilde{s}_j \tag{21}$$

Here, β controls local/communication information weighting, and ω_{ij} represents neighbor j's importance to agent i. The fuzzy communication message is defined as:

$$C_n^F(t) = \sum_{m \in \mathcal{N}_n} W_{n,m} O_m^F(t) \tag{22}$$

where $\mathcal{N}_n$ denotes agent n's communication neighborhood, and $W_{n,m}$ represents dynamic importance weights adjusted through historical interactions. The final input state becomes:

$$S_n(t) = \alpha O_n^F(t) + (1-\alpha) C_n^F(t) \tag{23}$$

with $\alpha \in [0,1]$ balancing local and global information.

To maximize user service hit rewards, agent n's immediate reward at timestep t is defined as:

$$R_n(t) = \sum_{f \in F} \left[\left(\omega Q_f(t) - \mu L_f(t)\right) \cdot X_f(t) \cdot \left(1 - U_f(t)\right) \right] \tag{24}$$

where $Q_f(t)$ denotes request count, $L_f(t)$ link latency, $X_f(t)$ backup status, and $U_f(t)$ failure rate for service chain f.

Popularity and failure rates are dynamically updated using exponentially weighted moving averages:

$$q_f(t+1) = (1-\beta_q)\, q_f(t) + \beta_q Q_f(t) \tag{25}$$

$$u_f(t+1) = (1-\beta_u)\, u_f(t) + \beta_u U_f(t) \tag{26}$$

Reward normalization employs fuzzy membership functions:

$$\mu_R(R_n) = \begin{cases} 1, & R_n < R_{\min} \\ \frac{R_{\max} - R_n}{R_{\max} - R_{\min}}, & R_{\min} \leq R_n \leq R_{\max} \\ 0, & R_n > R_{\max} \end{cases} \tag{27}$$

Similarly, request intensity and latency membership functions are:

$$\begin{aligned} \mu_Q\left(Q_f\right) &= e^{-\lambda_Q\left|Q_f-\overline{Q_f}\right|} \\ \mu_L\left(L_f\right) &= 1-e^{-\lambda_L\left(L_f-\overline{L_f}\right)} \end{aligned} \tag{28}$$

The fuzzy reward function becomes:

$$\tilde{R}_n(t)=\mu_R\left(R_n\right) \cdot \mu_Q\left(Q_j\right) \cdot\left(1-\mu_L\left(L_f\right)\right) \tag{29}$$

where the fuzzy reward $\tilde{R}_n(t)$ is the actual scalar signal used in the policy gradient update. The linear reward $R_n(t)$ is first computed based on service metrics and then transformed through fuzzy membership functions to obtain the final training reward. To ensure consistency between reward evaluation and policy optimization, the fuzzy reward $\tilde{R}_n(t)$ is directly adopted as the return signal in gradient computation.

Policy optimization uses fuzzy policy gradient methods:

$$\nabla_\theta J\left(\pi_n\right)=\mathbb{E}_{\pi_n}\left[\sum_{t=0}^T \tilde{R}_n(t) \nabla_\theta \log \pi_n\left(a_n \mid S_n\right)\right] \tag{30}$$

The action of each agent n at time t consists of selecting a backup deployment decision among its feasible neighboring edge nodes. Formally, let $\mathcal{N}_n(t)$ denote the dynamic neighbor set determined by the current network topology and resource availability. The discrete action space at time t is defined as:

$$\mathcal{A}_n(t)=\{a_{n,1}, a_{n,2}, \ldots, a_{n,|\mathcal{N}_n(t)|}\}, \tag{31}$$

where each action corresponds to selecting one candidate neighbor node for service function chain backup placement.

To accommodate the time-varying cardinality $|\mathcal{N}_n(t)|$, the policy network maintains a fixed maximum output dimension $A_{\max}$, which corresponds to the maximum possible neighbor size in the network. At each timestep, a binary action mask $M_n(t) \in \{0,1\}^{A_{\max}}$ is constructed such that:

$$M_{n,i}(t)=\begin{cases}1, & \text{if } i \in \mathcal{N}_n(t), \\ 0, & \text{otherwise.}\end{cases} \tag{32}$$

The masked policy distribution is then defined as:

$$\pi_n(a_i \mid S_n)=\frac{\exp(z_i) M_{n,i}(t)}{\sum_{j=1}^{A_{\max}} \exp(z_j) M_{n,j}(t)}, \tag{33}$$

where z_i denotes the unnormalized logit produced by the policy network.

This masked softmax mechanism ensures that invalid actions (corresponding to non-neighbor nodes) receive zero probability while preserving a fixed network output dimension. Therefore, the model remains scalable under dynamic topology changes without requiring structural modifications of the neural network.

Theorem 2 (Universal Approximation of Fuzzy Systems): Let $f: X \rightarrow \mathbb{R}$ be continuous. For any $\varepsilon>0$, there exists a fuzzy inference system $F(x)$ satisfying $\sup_{x \in X}|F(x)-f(x)|<\varepsilon$ [41].

Proof: By the Stone-Weierstrass theorem, $f(x)$ can be approximated by:

$$f(x) \approx \sum_{i=1}^N w_i \varphi_i(x) \tag{34}$$

Using fuzzy membership functions $\mu_i(x)$ as basis functions:

$$F(x)=\sum_{i=1}^N w_i \mu_i(x) \tag{35}$$

Normalization ($\sum_{i=1}^N \mu_i(x)=1$) and continuity ensure ε can be arbitrarily small.

Theorem 3 (Noise Suppression in Fuzzy Inference) [42]: Let s be the true signal and n additive Gaussian noise. Processing observation $y=s+n$ through fuzzy inference yields:

$$\mathbb{E}[F(y)]=s, \quad \operatorname{Var}[F(y)]<\operatorname{Var}[n] \tag{36}$$

Proof: Define fuzzy mapping:

$$F(y)=\sum_{i=1}^N w_i \mu_i(y) \tag{37}$$

For $n \sim \mathcal{N}\left(0, \sigma_n^2\right)$:

$$\mathbb{E}[F(y)]=\sum_{i=1}^N w_i \mu_i(s)=s \tag{38}$$

Variance reduction stems from:

$$\operatorname{Var}[F(y)]=\sum_{i=1}^N w_i^2 \operatorname{Var}\left[\mu_i(y)\right]<\operatorname{Var}[n] \tag{39}$$

as fuzzy operators suppress high-frequency noise.

Theorem 4 (Communication Overhead Reduction under Fuzzy Abstraction): Consider a multi-agent system where each agent transmits its observed state to neighbors. Let the raw numerical state be an n-dimensional vector $s \in \mathbb{R}^n$, and each scalar element be encoded using b bits. The raw communication overhead per message is [43]:

$$C_{\text{raw}}=bn. \tag{40}$$

Assume the fuzzy inference system maps s to an m-dimensional membership vector $O^F \in \mathbb{R}^m$, where $m<n$, and each membership value is also encoded with b bits. The fuzzy communication overhead per message becomes:

$$C_{\text{fuzzy}}=bm. \tag{41}$$

If fuzzy abstraction additionally reduces communication frequency from N_{msg} to N_{msg}' due to event-triggered or confidence-based transmission (i.e., $N_{\text{msg}}' \leq N_{\text{msg}}$), then the total communication cost over a time horizon T satisfies:

$$\frac{C_{\text{fuzzy}}^{\text{total}}}{C_{\text{raw}}^{\text{total}}}=\frac{N_{\text{msg}}' \, m}{N_{\text{msg}} \, n}. \tag{42}$$

Therefore, if

$$m<n \quad \text{and/or} \quad N_{\text{msg}}'<N_{\text{msg}}, \tag{43}$$

This article has been accepted for publication in IEEE Transactions on Mobile Computing. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TMC.2026.3688952

the fuzzy communication mechanism strictly reduces total communication overhead.

Proof: For raw state transmission, the total communication overhead over horizon T is:

$$C_{\text{raw}}^{\text{total}} = N_{\text{msg}} \cdot bn. \tag{44}$$

Under fuzzy abstraction, each transmitted message contains m membership values:

$$C_{\text{fuzzy}}^{\text{total}} = N'_{\text{msg}} \cdot bm. \tag{45}$$

Taking their ratio yields:

$$\frac{C_{\text{fuzzy}}^{\text{total}}}{C_{\text{raw}}^{\text{total}}} = \frac{N'_{\text{msg}} bm}{N_{\text{msg}} bn} = \frac{N'_{\text{msg}}}{N_{\text{msg}}} \cdot \frac{m}{n}. \tag{46}$$

Since b cancels, the reduction factor depends only on dimensionality compression and transmission frequency reduction [44].

Case 1 (Dimensionality Reduction): If $m < n$ and $N'_{\text{msg}} = N_{\text{msg}}$, then

$$\frac{C_{\text{fuzzy}}^{\text{total}}}{C_{\text{raw}}^{\text{total}}} = \frac{m}{n} < 1. \tag{47}$$

Case 2 (Frequency Reduction): If $m = n$ but $N'_{\text{msg}} < N_{\text{msg}}$, then

$$\frac{C_{\text{fuzzy}}^{\text{total}}}{C_{\text{raw}}^{\text{total}}} = \frac{N'_{\text{msg}}}{N_{\text{msg}}} < 1. \tag{48}$$

Case 3 (Combined Effect): If both $m < n$ and $N'_{\text{msg}} < N_{\text{msg}}$, then the communication reduction factor is multiplicative:

$$\frac{C_{\text{fuzzy}}^{\text{total}}}{C_{\text{raw}}^{\text{total}}} < \frac{m}{n}. \tag{49}$$

Hence, fuzzy abstraction provides a provable upper bound on communication overhead reduction proportional to $\frac{m}{n}$ and $\frac{N'_{\text{msg}}}{N_{\text{msg}}}$.

To address the dynamic and failure-aware SFC backup problem, Algorithm 1 formulates a structured multi-agent fuzzy reinforcement learning procedure.

Lines 1–4 initialize the learnable policy parameters, fuzzy membership functions, fuzzy rule base, and the experience replay buffer, which together constitute the learning foundation of the proposed framework. For each training episode (Line 5), every agent interacts with its local environment (Line 6) by observing the system state composed of resource availability, latency, service popularity, and failure rates. The observed state is transformed into a fuzzy representation through predefined membership functions (Line 7), enabling uncertainty modeling and dimensional abstraction. In Line 8, fuzzy inference is conducted via the rule base to generate high-level decision-support variables. These inference outputs are then used to determine whether a backup operation should be triggered (Line 9).

If a backup is required, the agent samples an action from the current stochastic policy (Line 11). Otherwise, the previous deployment strategy is maintained (Line 13), ensuring stability in non-critical conditions. After executing the selected action (Line 15), the agent observes the subsequent state (Line 16) and computes a reward signal (Line 17), which integrates service popularity, latency performance, and failure risk in a weighted manner. The transition tuple is appended to the experience buffer (Line 18), and the policy parameters are updated using gradient-based reinforcement learning (Line 19). This iterative process continues until convergence, yielding the optimized backup policy returned in Line 22.

Algorithm 1 Latency-Failure-Aware Multi-Agent Fuzzy Reinforcement Learning (LFA-MAFRL)

Require: Edge servers N, links L, SFC set S, user requests U, rsource capacity R_n, latency d_{ij}, failure rate F_n, popularity P_s
Ensure: Optimized backup policy π_θ
1: Initialize policy parameters θ
2: Initialize fuzzy membership functions $\mu(\cdot)$
3: Initialize fuzzy rule base $\mathcal{R}$
4: Initialize experience buffer $\mathcal{D}$
5: **for** episode $= 1$ to E **do**
6:     **for** each agent $a \in \mathcal{A}$ **do**
7:         Observe state $s_t^a \leftarrow \{R_n, d_{ij}, P_s, F_n\}$
8:         Compute fuzzy representation: $\tilde{s}_t^a \leftarrow \mu(s_t^a)$
9:         Perform fuzzy inference: $z_t^a \leftarrow \textsc{FuzzyInference}(\tilde{s}_t^a, \mathcal{R})$
10:         Determine backup requirement: $b_t^a \leftarrow \textsc{BackupDecision}(z_t^a)$
11:         **if** $b_t^a = 1$ **then**
12:             Select action $a_t^a \sim \pi_\theta(\tilde{s}_t^a)$
13:         **else**
14:             $a_t^a \leftarrow a_{t-1}^a$
15:         **end if**
16:         Execute action a_t^a
17:         Observe next state s_{t+1}^a
18:         Compute reward: $r_t^a \leftarrow \lambda_1 P_s + \lambda_2(1 - d_{ij}) - \lambda_3 F_n$
19:         Store transition: $\mathcal{D} \leftarrow \mathcal{D} \cup \{s_t^a, a_t^a, r_t^a, s_{t+1}^a\}$
20:         Update parameters: $\theta \leftarrow \theta + \alpha \nabla_\theta J(\theta)$
21:     **end for**
22: **end for**
23: **return** π_θ

IV. Experiments and Analysis

A. Experimental Settings

The experiments are conducted on the open-source simulation platform SFCSim [45], which enables the construction of a multi-access edge computing (MEC) environment. This platform supports dynamic network topology modeling, resource status monitoring, and full life-cycle management of Service Function Chains (SFCs). The simulation environment consists of 50 to 100 edge server nodes, where the link delay between nodes follows a uniform distribution within the range of [5ms, 50ms], and the storage capacity of each node is configured between [20GB, 100GB]. User requests arrive according to a Poisson process, with 100 to 200 mobile terminals in the system. Terminal mobility follows a random walk model with average speeds ranging from 5 to 15m/s. The SFC set includes 10 different types, each composed of 3 to 8 sequential Virtual Network Functions (VNFs), where each VNF demands between 1GB and 5GB of resources. The parameter settings are referenced from [9], [11], [45].

The proposed approach is compared with three state-of-the-art baselines. The first is an online Bandit learning scheme (denoted as **Bandit**) proposed by Wang et al. [11], which estimates the popularity and failure rate of SFCs using a multi-armed bandit model and applies a Prim-based heuristic to optimize link latency. The second is a deep reinforcement learning-based adaptive dynamic migration optimization algorithm (referred to as **DRL-ADMO**) proposed by Xu et al. [9], which formulates the SFC migration task as an integer programming problem and utilizes deep reinforcement learning to determine efficient migration paths and bandwidth allocation, enabling seamless parallel migration. The third is a fuzzy-assisted dueling double deep Q-network with regularization algorithm (referred to as **D3QNReg**) proposed by Wu et al. [37], which incorporates fuzzy QoS–QoE modeling into the MDP-based SFC orchestration process to improve learning stability and adaptive deployment. Our method is mentioned as **LFA-MAFRL**.

To thoroughly evaluate the adaptability and robustness of the proposed method, three representative experimental scenarios are designed to reflect key challenges in dynamic edge environments. The first involves *random node failures*, where edge servers fail unpredictably with a probability drawn from a Bernoulli distribution in the range [0.01, 0.1]. The second considers *user mobility*, where mobile terminals continuously update their access points based on an enhanced random walk model, capturing frequent handovers among edge nodes. The third simulates *varying workloads* by adjusting the request arrival rates to represent low, medium, and high levels of system load, thus examining the responsiveness and scalability of the proposed strategy under different resource competition conditions.

For performance evaluation, four key metrics are selected to systematically compare the proposed method with the baselines. Service Hit Reward measures the accumulated reward obtained when user requests are successfully served by the deployed SFC instances. This metric reflects the effectiveness of the backup strategy in maintaining service availability and satisfying user demand. Service Delay measures the average time from request initiation to service completion, indicating the method's ability to ensure QoS. Migration Success Rate represents the proportion of successfully completed SFC migrations, reflecting the system's robustness against network disturbances. Migration Time records the average time required to complete each migration, assessing the cost-efficiency of the method in maintaining service continuity while minimizing migration overhead.

B. Overall Performance

As shown in Figure 3, to provide an intuitive and comprehensive understanding of how different methods respond to increasing failure intensities, the actual backup schemes under three different failure rates ($p = 1\%$, 10%, and 15%) are visualized for each method—Bandit, DRL-ADMO, D3QNReg, and the proposed LFA-MAFRL. Each topology snapshot highlights active service nodes, backup paths, and user locations, where red nodes indicate failed servers and green nodes represent mobile users at the time of migration. As failure probability increases, the spatial distribution of backups becomes more dispersed and redundant to account for increased risk, clearly showing how the robustness mechanisms adapt to varying environmental stress.

In addition to the visual comparisons, the quantitative performance of all four methods across the three failure rates is reported using four key evaluation metrics: Service Delay (ms), Migration Success Rate (%), Migration Time (s), and Service Hit Reward. The results are summarized in Table III, which further verifies the consistency and generalizability of our proposed strategy across scenarios.

As shown in Table III, under all failure intensities, the proposed LFA-MAFRL method consistently achieves the lowest service delay and migration time while maintaining the highest migration success rate. The newly introduced D3QNReg method, which enhances policy stability through dueling architecture and regularization, demonstrates improved adaptability compared to DRL-ADMO, particularly under moderate and high failure rates. For instance, when the failure rate is low (1%), LFA-MAFRL reduces service delay by 4.9% compared to D3QNReg and by 8.3% compared to DRL-ADMO. Its migration success rate remains 2.3% higher than D3QNReg and 4.5% higher than DRL-ADMO. As the failure rate increases to 15%, the performance gap remains evident, with LFA-MAFRL achieving a delay reduction of 3.9% over D3QNReg and maintaining a success rate improvement of 3.1%.

These results reflect the resilience and adaptability of the LFA-MAFRL framework under stress. While Bandit and DRL-ADMO exhibit noticeable degradation in both latency and reliability as node failures become more severe, D3QNReg mitigates part of this decline through improved training stability and adaptive decision-making. Nevertheless, LFA-MAFRL's proactive failure-aware design and fuzzy coordination mechanisms enable more effective backup allocation and smoother migration execution.

The advantage of LFA-MAFRL is further confirmed in terms of Service Hit Reward. It consistently achieves the highest reward across all failure rates, indicating that its successful migrations translate into effective service availability rather than merely maintaining nominal connectivity. Under low failure intensity (1%), LFA-MAFRL improves the reward by 5.4% over D3QNReg and 9.9% over DRL-ADMO. When the failure rate increases to 15%, the improvement reaches 9.3% and 18.0%, respectively. This demonstrates that while D3QNReg enhances learning stability and resource utilization, LFA-MAFRL more effectively preserves end-to-end service usability under dynamic failures by jointly consid-

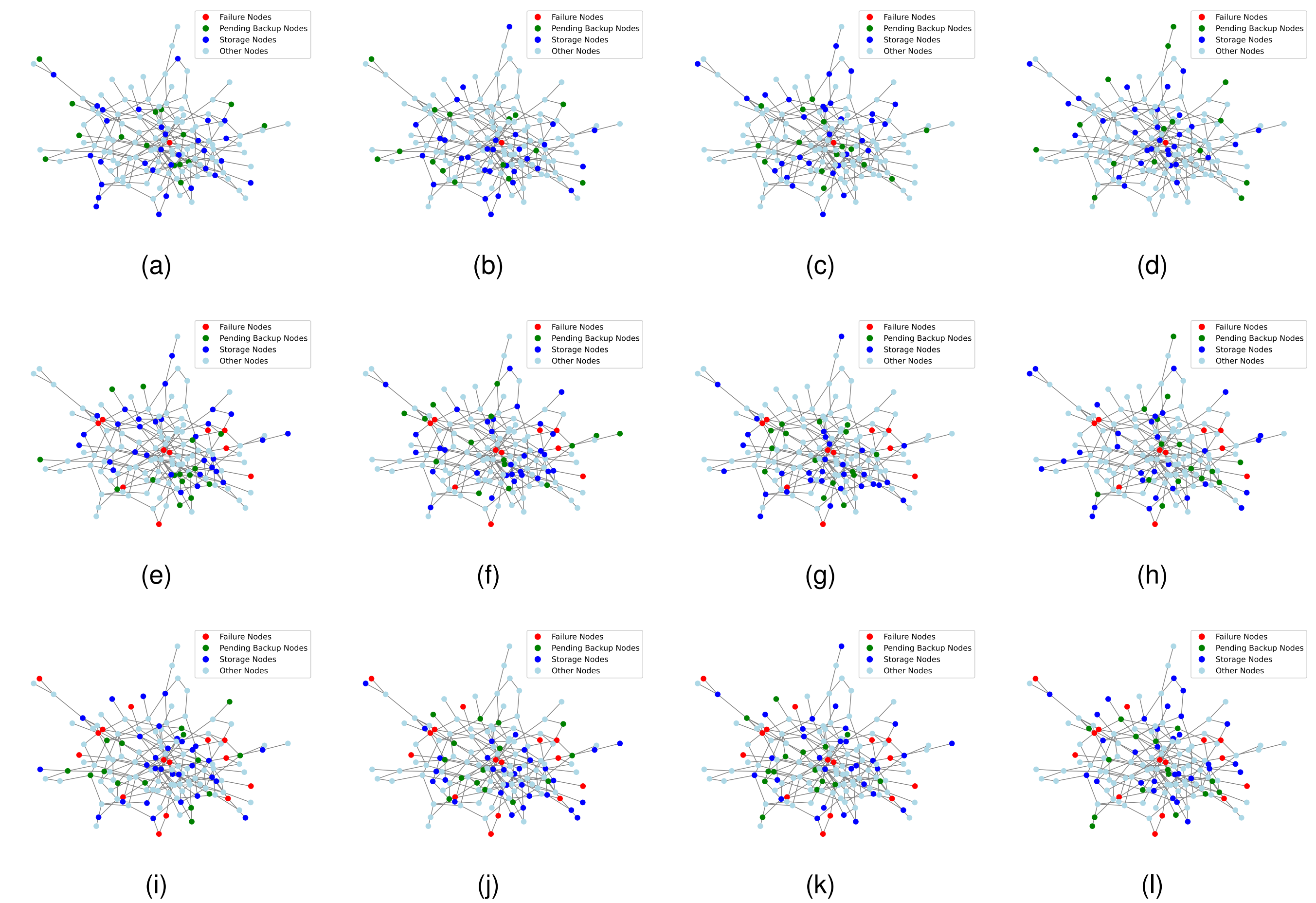


Fig. 3: Data Distribution Solution in Overall Performance. (a) Brandit with 1% Failure Rate; (b) DRL-ADMO with 1% Failure Rate; (c) D3QNReg with 1% Failure Rate; (d)LFA-MAFRL with 1% Failure Rate; (e) Brandit with 10% Failure Rate; (f) DRL-ADMO with 10% Failure Rate; (g) D3QNReg with 10% Failure Rate; (h) LFA-MAFRL with 10% Failure Rate; (i) Brandit with 15% Failure Rate; (j) DRL-ADMO with 15% Failure Rate; (k) D3QNReg with 15% Failure Rate; (l) LFA-MAFRL with 15% Failure Rate.

ering latency-awareness and fuzzy coordination in its decision process.

C. Random Node Failures

As shown in Figure 4, to evaluate the robustness of different methods under unexpected infrastructure disruptions, random node failures are simulated in a dynamic edge computing environment consisting of 50 to 100 nodes. Node failures follow a Weibull distribution with a probability $p \in [0.01, 0.1]$, and failure durations are drawn from an exponential distribution with $\lambda = 1/30$. The system processes a fixed request rate of 80 requests per second.

Experimental results indicate that the proposed LFA-MAFRL method consistently achieves the best performance across all evaluation metrics, while the newly introduced D3QNReg demonstrates improved robustness compared to DRL-ADMO. In terms of service delay, LFA-MAFRL achieves an average latency of 98–119,ms, which is 7–9% lower than DRL-ADMO and up to 17% lower than Bandit. D3QNReg shows noticeable improvement over DRL-ADMO due to its stabilized value estimation and regularization-enhanced learning process, while remaining slightly inferior to LFA-MAFRL under severe failures.

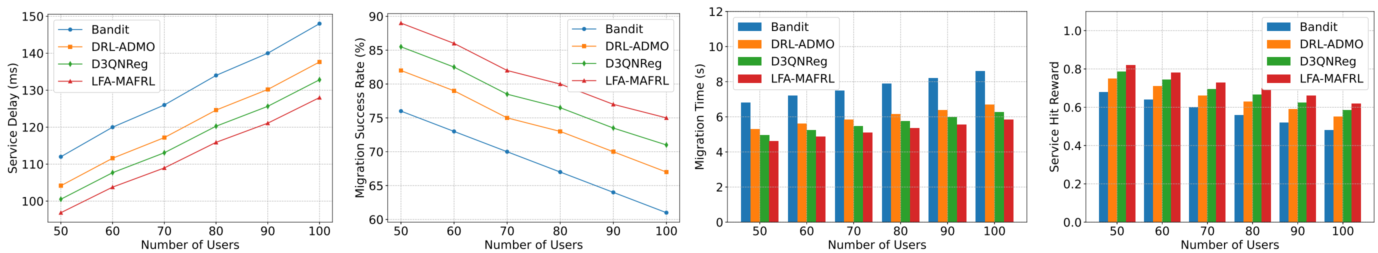
For migration success rate, LFA-MAFRL maintains a stable performance of 75–89%, outperforming DRL-ADMO's 67–82%. D3QNReg achieves intermediate performance by improving policy stability under dynamic node failures, reducing the performance gap between DRL-based approaches and the fuzzy-coordinated LFA-MAFRL.

Meanwhile, average migration time is reduced to 4.5–5.9s under LFA-MAFRL, compared to 5.2–6.8s for DRL-ADMO and 6.8–8.6s for Bandit. D3QNReg further improves migration efficiency compared to DRL-ADMO by mitigating oscillatory decision behaviors during backup placement, yet still requires slightly longer time than LFA-MAFRL due to its lack of proactive failure anticipation.

In addition to these migration-centric metrics, the Service Hit Reward further reflects the effective service availability perceived by users under dynamic failures. Results show that LFA-MAFRL consistently achieves the highest Service Hit

TABLE III: Overall Performance Comparison Under Varying Failure Rates

Method	Service Delay (ms)	Migration Success Rate (%)	Migration Time (s)	Service Hit Reward
Failure Rate = 1%				
Bandit	108.2	76.5	6.7	0.62
DRL-ADMO	99.3	84.1	5.9	0.71
D3QNReg	95.8	86.3	5.6	0.74
LFA-MAFRL	91.1	88.6	5.3	0.78
Failure Rate = 10%				
Bandit	134.7	68.9	7.9	0.49
DRL-ADMO	122.5	76.4	6.8	0.58
D3QNReg	118.4	79.1	6.5	0.61
LFA-MAFRL	112.1	82.5	6.2	0.66
Failure Rate = 15%				
Bandit	149.6	61.3	8.3	0.41
DRL-ADMO	135.4	70.2	7.5	0.50
D3QNReg	131.2	73.4	7.1	0.54
LFA-MAFRL	126.1	76.5	6.9	0.59



(a) (b) (c) (d)

Fig. 4: Random Node Failure Results. (a) Service Delay; (b) Migration Success Rate; (c) Migration Time; (d) Service Hit Reward.

Reward across all network scales. D3QNReg provides moderate improvements over DRL-ADMO by enhancing training stability and deployment consistency, but its reactive decision-making still limits service usability under high node failure intensity.

The superior performance of LFA-MAFRL stems from its proactive failure-aware design. Unlike Bandit and DRL-ADMO, and compared to the stability-driven D3QNReg, LFA-MAFRL integrates predictive modeling to anticipate potential service disruptions and initiates timely backup migrations. Furthermore, its incremental synchronization mechanism minimizes the overhead of state transfer during migration, thereby reducing service interruption while preserving consistency. These features enable LFA-MAFRL to achieve a favorable trade-off between responsiveness and stability under dynamic and failure-prone network conditions.

D. User Mobility

As shown in Figure 5, in the user mobility scenario, a dynamic environment is simulated where 100 to 200 mobile terminals switch access points at speeds ranging from 5 to $15\mathrm{m/s}$, evaluating the impact of user mobility on service continuity.

Experimental results show that the Bandit approach suffers from significant service delay increases due to link jitter caused by frequent handovers, with delays ranging from $138\mathrm{ms}$ to $168\mathrm{ms}$. The DRL-ADMO method mitigates part of the delay by employing path preallocation, reducing latency to the range of $126\mathrm{ms}$–$153\mathrm{ms}$. The newly introduced D3QNReg further improves delay stability through its dueling architecture and regularized value updates, achieving moderate latency reduction compared to DRL-ADMO while remaining slightly inferior to the predictive LFA-MAFRL.

In contrast, the proposed LFA-MAFRL method leverages trajectory prediction to anticipate access point switching and proactively reserve computational and network resources. As a result, the service delay is reduced to $115\mathrm{ms}$, $126\mathrm{ms}$, and $135\mathrm{ms}$ across different user scales, achieving an additional 8.7% reduction compared to DRL-ADMO. These findings demonstrate LFA-MAFRL's superior adaptability to dynamic topological changes induced by user mobility, while D3QNReg provides a more stable intermediate solution.

In terms of migration success rate, the Bandit method achieves only 56% to 71% due to frequent handover conflicts and lack of fault tolerance. DRL-ADMO improves this metric to 64%–79% through reinforcement learning-based migration control. D3QNReg further enhances migration reliability by stabilizing policy learning under mobility-induced uncertainty, achieving performance between DRL-ADMO and LFA-MAFRL. The latter outperforms both by employing a preemptive resource reservation mechanism, maintaining a

This article has been accepted for publication in IEEE Transactions on Mobile Computing. This is the author's version which has not been fully edited and
content may change prior to final publication. Citation information: DOI 10.1109/TMC.2026.3688952

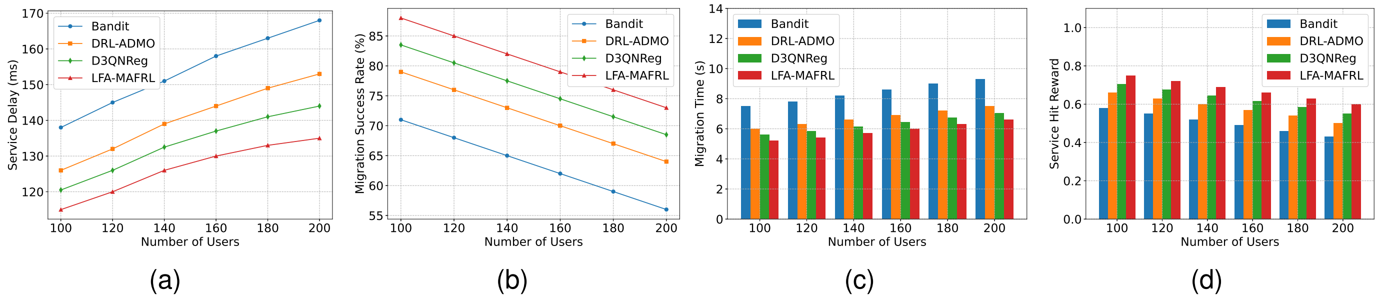


Fig. 5: User Mobility Results. (a) Service Delay; (b) Migration Success Rate; (c) Migration Time; (d) Service Hit Reward.

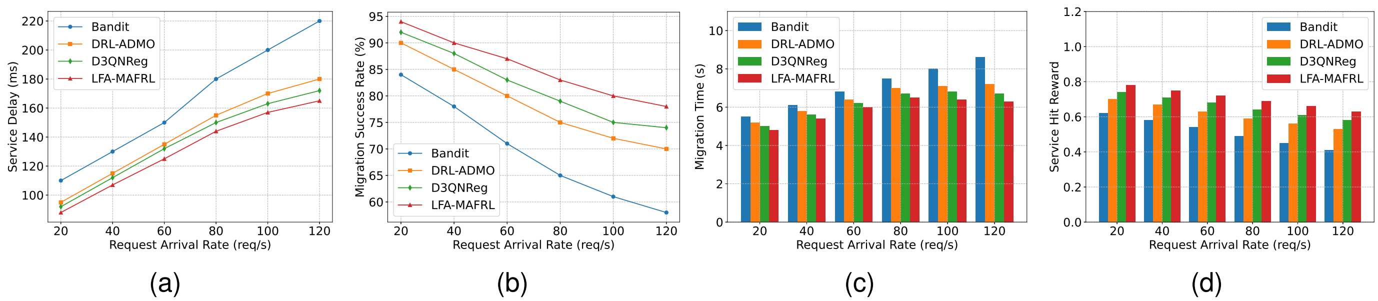


Fig. 6: Workload-Aware Scalability Evaluation Results. (a) Service Delay; (b) Migration Success Rate; (c) Migration Time; (d) Service Hit Reward.

stable success rate between 88% and 73%.

Moreover, LFA-MAFRL demonstrates reduced migration time (5.2s to 6.6s) compared to DRL-ADMO (6.0s to 7.5s). D3QNReg again provides intermediate performance by reducing oscillatory migrations while lacking the fully proactive synchronization capability of LFA-MAFRL.

A similar performance hierarchy is reflected in the Service Hit Reward. LFA-MAFRL consistently maintains the highest effective service availability across all mobility levels, followed by D3QNReg and DRL-ADMO. As user mobility intensifies, baseline methods experience noticeable degradation in service utility despite moderate migration success, whereas LFA-MAFRL preserves stable service delivery through proactive resource preparation and trajectory-aware backup placement. This confirms LFA-MAFRL's effectiveness in ensuring fast, reliable, and usable service migration in highly dynamic mobile edge computing environments.

E. Workload-Aware Scalability Evaluation

As shown in Figure 6, to assess scalability under varying system pressure, request arrival rates ranging from 20 to 120 requests per second are simulated to emulate light, medium, and heavy workloads. Under heavy load (120 req/s), the baseline Bandit approach suffers from a significant increase in service delay, reaching up to 220ms due to inefficient resource contention handling. DRL-ADMO mitigates this by learning resource allocation policies, reducing delay to 180,ms. The D3QNReg method further improves delay performance through more stable value estimation and regularized decision updates, achieving a delay of approximately 172ms under heavy workload. Our proposed LFA-MAFRL framework further lowers the delay to 165,ms by introducing load-aware proactive scheduling, resulting in the best latency performance across all workload levels.

In terms of migration success rate, Bandit performs poorly under stress, dropping to 58%. DRL-ADMO achieves moderate stability with 70%, while D3QNReg enhances decision consistency and improves the success rate to around 74% under high load conditions. LFA-MAFRL maintains the highest success rate of 78% through anticipatory resource reservation. Although migration time increases across all methods under pressure, LFA-MAFRL consistently maintains the lowest duration (6.3s compared to 7.2s for DRL-ADMO), while D3QNReg achieves intermediate performance at approximately 6.7s, demonstrating improved responsiveness over DRL-ADMO.

Additionally, Service Hit Reward is evaluated to reflect the system's ability to sustain effective service provisioning under dynamic workloads. As workload intensity increases, Bandit exhibits a noticeable degradation in reward due to reactive and myopic decision-making. DRL-ADMO demonstrates improved robustness by leveraging learned allocation strategies. D3QNReg further enhances reward stability through value regularization and improved policy convergence. In contrast, LFA-MAFRL consistently achieves the highest reward across all workload levels, indicating that its load-aware foresight and fuzzy decision guidance enable more stable and efficient service matching under resource pressure.

This article has been accepted for publication in IEEE Transactions on Mobile Computing. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TMC.2026.3688952

F. Ablation Study on LFA-MAFRL

To rigorously validate the effectiveness of the fuzzy logic integration within the proposed LFA-MAFRL framework, an ablation study is conducted by progressively removing key components related to fuzzy objective transformation and fuzzy multi-agent communication.

As shown in Table IV, two main fuzzy-related modules are removed to construct degraded variants: the fuzzy objective transformation module and the fuzzy communication mechanism within the multi-agent reinforcement learning process.

The fuzzy objective transformation module is responsible for converting the conflicting objectives of service hit reward maximization and latency minimization into a unified fuzzy optimization framework using membership functions and the ε-constraint method. When this module is removed, the model relies on fixed reward structures, which weakens its ability to balance trade-offs under dynamic workloads. As a result, both service delay and reward performance degrade.

The fuzzy communication mechanism enhances inter-agent coordination by filtering noisy local observations through fuzzy inference before information exchange. Removing this module leads to increased uncertainty in decentralized decision-making, which manifests as reduced migration success rates and longer migration time due to suboptimal backup placement decisions.

When both fuzzy modules are removed, the system degenerates into a standard MARL architecture. In this case, performance drops significantly across all metrics, confirming that the improvements achieved by LFA-MAFRL are not merely due to reinforcement learning itself, but are largely attributable to the integration of fuzzy reasoning.

Overall, the complete LFA-MAFRL achieves the best performance by jointly leveraging fuzzy objective normalization and noise-resilient multi-agent coordination. The former ensures adaptive multi-objective balancing, while the latter improves robustness in partially observable environments, ultimately leading to lower delay, higher migration reliability, reduced migration overhead, and improved service provisioning effectiveness.

To further verify that the proposed framework can effectively learn a policy through continuous interaction with the environment, we provide the convergence curves of the average cumulative reward during training, as illustrated in Figure 7. The horizontal axis represents the training episodes, while the vertical axis denotes the average episode reward obtained by the agents.

It can be observed that all methods exhibit a clear upward trend as training progresses, indicating that the agents gradually improve their decision-making policies through reinforcement learning. In particular, the proposed full LFA-MAFRL framework achieves the fastest convergence speed and the highest stable reward, demonstrating the effectiveness of integrating fuzzy objective transformation and fuzzy communication mechanisms into the multi-agent reinforcement learning process.

Furthermore, although fuzzy rules are used to guide the initial decision space, the continuous increase in cumulative reward throughout the training episodes indicates that the final

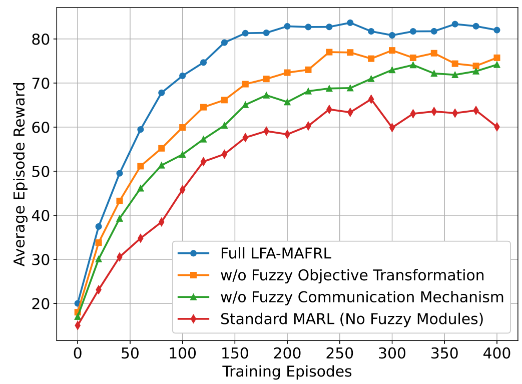


Fig. 7: Training Convergence Comparison.

policy is not solely determined by the initialized fuzzy knowledge. Instead, the agents progressively refine their strategies through trial-and-error interactions with the environment. In contrast, removing the fuzzy objective transformation or the fuzzy communication mechanism leads to slower convergence and lower final rewards. The standard MARL approach without fuzzy modules shows the slowest convergence and the lowest performance due to the lack of structured guidance during learning.

These results demonstrate that the fuzzy components serve as auxiliary mechanisms that facilitate more efficient exploration and coordination, while the final policy is still learned through reinforcement learning optimization.

V. Practical Deployment, Limitations, and Computational Complexity

A. Deployment Challenges in Real Environments

Deploying the proposed LFA-MAFRL framework in real MEC environments faces several practical challenges. Timely collection of service request and failure data is required, and edge servers must communicate their local states efficiently. Network delays, partial observability, or communication loss may degrade the performance of the learned policies. Moreover, tuning hyperparameters such as fuzzy membership functions and ε-constraint thresholds is necessary to adapt to different operational conditions. Despite these challenges, the fuzzy communication mechanism helps mitigate observation noise, while the decentralized multi-agent design reduces the need for global coordination, facilitating feasible real-world deployment.

B. Limitations and Constraints

While LFA-MAFRL achieves strong adaptive performance, several limitations exist. First, the algorithm assumes sufficient historical data to estimate request popularity and failure rates; in highly sparse or bursty scenarios, performance may be affected. Second, the action space is designed for local decision-making and may not fully capture long-range dependencies across all nodes. Third, extreme network scales or very high update frequencies could increase both computation and communication overhead, potentially delaying decisions. Finally,

TABLE IV: Ablation Study Results of LFA-MAFRL

Method	Service Delay (ms)	Migration Success Rate (%)	Migration Time (s)	Service Hit Reward
Full LFA-MAFRL	165	78	6.3	0.63
w/o Fuzzy Objective Transformation	173	74	6.8	0.58
w/o Fuzzy Communication Mechanism	179	72	7.0	0.55
Standard MARL (No Fuzzy Modules)	188	69	7.4	0.51

the method provides near-optimal solutions under dynamic stochastic conditions, but does not guarantee globally optimal SFC backup allocations due to the NP-hard nature of the problem.

C. Computational Complexity Analysis

The computational complexity of LFA-MAFRL per agent at each timestep involves fuzzy inference, local state fusion, and policy evaluation. Let $|N_i|$ denote the number of neighbors for agent i, m the number of fuzzy sets, and $|\mathcal{A}_i|$ the action space size. The per-agent complexity can be expressed as:

$$\mathcal{O}\big(|N_i| \cdot m + |\mathcal{A}_i|\big) \tag{50}$$

This ensures scalability in large-scale networks, as each agent only performs local computations and exchanges compact fuzzy messages. The fuzzy communication mechanism reduces high-dimensional state sharing, further limiting communication overhead. However, in extremely large or highly dynamic networks, the cumulative complexity may introduce delays, which should be considered in real-time deployments.

VI. Conclusion

This paper proposes a Latency-Failure-Aware Multi-Agent Fuzzy Reinforcement Learning (LFA-MAFRL) for service function chain (SFC) backup in dynamic and failure-prone MEC environments. The proposed method effectively balances service availability and network latency under uncertain conditions. Through the use of ε-constrained fuzzy optimization and fuzzy inference-driven agent communication, the approach addresses key limitations of existing solutions, such as limited adaptability and poor decision quality in partially observable environments. Extensive experimental evaluations on the SFC-Sim platform demonstrate that the framework significantly improves SFC reliability and responsiveness compared to state-of-the-art baselines.

Future work will focus on extending the model in two directions. First, user behavior prediction and demand forecasting will be incorporated to enhance the proactivity of SFC backup strategies. Second, trust-aware mechanisms and game-theoretic learning will be explored to address adversarial behaviors and incentive compatibility in multi-agent cooperation. Additionally, the approach will be tested in real-world MEC testbeds to evaluate its scalability and robustness under practical network conditions.

References

[1] L. Li, T. Q. Quek, J. Ren, H. H. Yang, Z. Chen, and Y. Zhang, "An incentive-aware job offloading control framework for multi-access edge computing," *IEEE Transactions on Mobile Computing*, vol. 20, no. 1, pp. 63–75, 2019.

[2] P. Cruz, N. Achir, and A. C. Viana, "On the edge of the deployment: A survey on multi-access edge computing," *ACM Computing Surveys*, vol. 55, no. 5, pp. 1–34, 2022.

[3] D. H. Nguyen, C.-C. Lin, T. N. Nguyen, S.-I. Chu, and B.-H. Liu, "Service recovery in nfv-enabled networks: Algorithm design and analysis," *IEEE Transactions on Cloud Computing*, vol. 12, no. 2, pp. 800–813, 2024.

[4] Y. Liu and J. Zhang, "Service function chain embedding meets machine learning: Deep reinforcement learning approach," *IEEE Transactions on Network and Service Management*, vol. 21, no. 3, pp. 3465–3481, 2024.

[5] P. A. Apostolopoulos, G. Fragkos, E. E. Tsiropoulou, and S. Papavassiliou, "Data offloading in uav-assisted multi-access edge computing systems under resource uncertainty," *IEEE Transactions on Mobile Computing*, vol. 22, no. 1, pp. 175–190, 2021.

[6] H. Biallach, M. Bouhtou, D. Nace, and M. S. Mimouna, "An efficient heuristic for the virtual network function reconfiguration problem," in *2022 12th International Workshop on Resilient Networks Design and Modeling (RNDM)*. IEEE, 2022, pp. 1–7.

[7] D. Li, Z. Wang, R. Zhao, H. Zhang, Z. Yin, N. Cheng, and J. Liu, "Dynamic sfc deployment for sdn/nfv-based satellite-terrestrial integrated vehicular networks," *IEEE Transactions on Vehicular Technology*, 2025.

[8] Y. Li, Q. Zhang, H. Yao, X. Xin, Y. Zhao, and R. Gao, "Optimizing cost-efficient sfc routing in ntns: a novel transformer-ant colony optimization framework," *IEEE Transactions on Vehicular Technology*, 2025.

[9] L. Xu, W. Liu, Z. Wang, J. Luo, J. Wang, and Z. Ma, "Mobile-aware service function chain intelligent seamless migration in multi-access edge computing," *Journal of Network and Systems Management*, vol. 32, no. 3, p. 49, 2024.

[10] S. Cherrared, S. Imadali, E. Fabre, and G. Gössler, "Sfc self-modeling and active diagnosis," *IEEE Transactions on Network and Service Management*, vol. 18, no. 3, pp. 2515–2530, 2021.

[11] C. Wang, Q. Hu, D. Yu, and X. Cheng, "Online learning for failure-aware edge backup of service function chains with the minimum latency," *IEEE/ACM Transactions on Networking*, vol. 31, no. 6, pp. 2730–2744, 2023.

[12] R. He, B. Ren, J. Xie, D. Guo, and L. Zhao, "Efficient online scheduling of service function chains across multiple geo-distributed regions," *IEEE Transactions on Network and Service Management*, 2024.

[13] K. Li, X. Wang, Q. He, J. Wang, J. Li, S. Zhan, G. Lu, and S. Dustdar, "Computation offloading in resource-constrained multi-access edge computing," *IEEE Transactions on Mobile Computing*, vol. 23, no. 11, pp. 10 665–10 677, 2024.

[14] M. Niu, Q. Han, B. Cheng, M. Wang, Z. Xu, W. Gu, S. Zhang, and J. Chen, "Hars: A high-available and resource-saving service function chain placement approach in data center networks," *IEEE Transactions on Network and Service Management*, vol. 19, no. 2, pp. 829–847, 2022.

[15] Y. Wang, L. Zhang, P. Yu, K. Chen, X. Qiu, L. Meng, M. Kadoch, and M. Cheriet, "Reliability-oriented and resource-efficient service function chain construction and backup," *IEEE Transactions on Network and Service Management*, vol. 18, no. 1, pp. 240–257, 2020.

[16] Y. Zhang, F. He, and E. Oki, "Availability-aware service provisioning with backup sub-chain-enabled sharing," in *GLOBECOM 2022-2022 IEEE Global Communications Conference*. IEEE, 2022, pp. 135–140.

[17] F. Yaghoubpour, B. Bakhshi, and F. Seifi, "End-to-end delay guaranteed service function chain deployment: A multi-level mapping approach," *Computer Communications*, vol. 194, pp. 433–445, 2022.

[18] F. Busacca, S. Palazzo, R. Raftopoulos, and G. Schembra, "Mantra: A distributed mab-based multi-agent framework for latency-and energy-aware offloading in vehicular networks," *IEEE Transactions on Network and Service Management*, 2025.

[19] I. M. Araújo, C. Natalino, H. Chen, M. De Andrade, D. L. Cardoso, and P. Monti, "Availability-guaranteed service function chain provisioning with optional shared backups," in *2020 16th International Conference on the Design of Reliable Communication Networks DRCN 2020*. IEEE, 2020, pp. 1–6.

[20] Q. Zhou, Y. Lian, J. Wu, M. Zhu, H. Wang, and J. Cao, "An optimized q-learning algorithm for mobile robot local path planning," *Knowledge-Based Systems*, vol. 286, p. 111400, 2024.

[21] Q. Zhou, J. Wu, M. Zhu, Y. Zhou, F. Xiao, and Y. Zhang, "Llm-ql: a llm-enhanced q-learning approach for scheduling multiple parallel drones," *IEEE Transactions on Knowledge and Data Engineering*, vol. 37, no. 9, pp. 5393–5406, 2025.

[22] H. Zeng, Z. Zhu, Y. Wang, Z. Xiang, and H. Gao, "Periodic collaboration and real-time dispatch using an actor–critic framework for uav movement in mobile edge computing," *IEEE Internet of Things Journal*, 2024.

[23] Q. Zhou, J. Wu, H. Dai, G. Yang, and Y. Zhang, "An intelligent ride-sharing recommendation method based on graph neural network and evolutionary computation," *IEEE Transactions on Intelligent Transportation Systems*, 2024.

[24] H. Byeon, M. Alsaadi, A. Quraishi, A. AlGhamdi, T. A. Ahanger, I. Keshta, P. A. Xalikovich, M. Soni, and M. W. Bhatt, "Consumer technology in task offloading and edge resource allocation: Aiot and edge computing for next-generation communication," *IEEE Transactions on Consumer Electronics*, 2025.

[25] B. Ren, Y. Tang, H. Wang, Y. Wang, J. Liu, G. Gao, and W. Wei, "A multi-agents deep reinforcement learning autonomous security management approach for internet of things," *IEEE Internet of Things Journal*, 2024.

[26] J. Cao, T. Zhou, S. Zhi, S. Lam, G. Ren, Y. Zhang, Y. Wang, Y. Dong, and J. Cai, "Fuzzy inference system with interpretable fuzzy rules: Advancing explainable artificial intelligence for disease diagnosis—a comprehensive review," *Information Sciences*, vol. 662, p. 120212, 2024.

[27] C. Zhu, D. Ye, T. Zhu, and W. Zhou, "The evolution of cooperation in continuous dilemmas via multi-agent reinforcement learning," *Knowledge-Based Systems*, vol. 315, p. 113153, 2025.

[28] T. Yao, Y. Xu, H. Wang, X. Qiu, K. Althoefer, and P. Qi, "Multi-agent fuzzy reinforcement learning with llm for cooperative navigation of endovascular robotics," *IEEE Transactions on Fuzzy Systems*, 2025.

[29] Y. Zhong, D. Zheng, and X. Cao, "A drl approach with network service deployment transformer for reliable sfc deployment," in *ICC 2024-IEEE International Conference on Communications*. IEEE, 2024, pp. 177–182.

[30] D. Niu, G. Yi, and Z. Tang, "Toward a resource-efficient service function chain mapping mechanism: A heuristic method," in *2024 IEEE 99th Vehicular Technology Conference (VTC2024-Spring)*. IEEE, 2024, pp. 1–5.

[31] X. Shang, Y. Huang, Z. Liu, and Y. Yang, "Reducing the service function chain backup cost over the edge and cloud by a self-adapting scheme," *IEEE Transactions on Mobile Computing*, vol. 21, no. 8, pp. 2994–3008, 2021.

[32] M. A. Madani, F. Zhou, and A. Meddahi, "Deploying disaster-resilient service function chains using adaptive multi-path routing," *IEEE Transactions on Network and Service Management*, 2024.

[33] A. Salari, S. E. Ahmadi, M. Marzband, and M. Zeinali, "Fuzzy q-learning-based approach for real-time energy management of home microgrids using cooperative multi-agent system," *Sustainable Cities and Society*, vol. 95, p. 104528, 2023.

[34] M. Zhang, Y. Dou, V. Marojevic, P. H. J. Chong, and H. C. Chan, "Faq: A fuzzy-logic-assisted q-learning model for resource allocation in 6g v2x," *IEEE Internet of Things Journal*, vol. 11, no. 2, pp. 2472–2489, 2023.

[35] D. Yang, S. Wang, Y. Liao, and R. Pan, "An online energy management strategy for fuel cell vehicles based on fuzzy q-learning and road condition recognition," *IEEE Transactions on Intelligent Transportation Systems*, 2024.

[36] S. Yu and C. Jiang, "A cloud computing virtual machine scheduling strategy based on fuzzy reinforcement learning," *Computer Engineering and Science*, vol. 1, 2025.

[37] Y. Wu, Z. Jia, Q. Wu, and Z. Lu, "Adaptive qoe-aware sfc orchestration in uav networks: A deep reinforcement learning approach," *IEEE Transactions on Network Science and Engineering*, vol. 11, no. 6, pp. 6052–6065, 2024.

[38] X. Ma, K. Zong, and A. Rezaeipanah, "Auto-scaling and computation offloading in edge/cloud computing: a fuzzy q-learning-based approach," *Wireless Networks*, vol. 30, no. 2, pp. 637–648, 2024.

[39] T. Q. Bao and B. S. Mordukhovich, "Sufficient conditions for global weak pareto solutions in multiobjective optimization," *Positivity*, vol. 16, no. 3, pp. 579–602, 2012.

[40] B. Fang, C. Zheng, H. Wang, and T. Yu, "Two-stream fused fuzzy deep neural network for multiagent learning," *IEEE Transactions on Fuzzy Systems*, vol. 31, no. 2, pp. 511–520, 2022.

[41] J. Lu, G. Ma, and G. Zhang, "Fuzzy machine learning: A comprehensive framework and systematic review," *IEEE Transactions on Fuzzy Systems*, 2024.

[42] L. Cunlin and Z. Qiang, "Nash equilibrium strategy for fuzzy non-cooperative games," *Fuzzy Sets and Systems*, vol. 176, no. 1, pp. 46–55, 2011.

[43] T. Wang, Q. Dang, and P. Pan, "A multi-robot system based on a hybrid communication approach," *Studies in Media and Communication*, vol. 1, no. 1, pp. 91–100, 2013.

[44] Z. Liu, M. Ang, and W. K. G. Seah, "Multi-robot concurrent learning of fuzzy rules for cooperation," in *2005 International Symposium on Computational Intelligence in Robotics and Automation*. IEEE, 2005, pp. 713–719.

[45] L. Xu, H. Hu, and Y. Liu, "Sfcsim: a network function virtualization resource allocation simulation platform," *Cluster Computing*, vol. 26, no. 1, pp. 423–436, 2023.

Qian Zhou (Member, IEEE) received her Ph.D. in Computer Application Technology from Nanjing University of Aeronautics and Astronautics (NUAA), China, in 2018. She is currently an Associate Professor at Nanjing University of Posts and Telecommunications (NJUPT), China, where she also serves as the Director of the System Security and Availability Engineering (SAVAGE) Application Technology Institute. Her research interests encompass system security and availability, applied cryptography, network security and privacy, Internet of Things (IoT) technologies, as well as database systems and other foundational theories and applications.

Jiayang Wu received the B.E. degree from Nanjing University of Posts and Telecommunications (NJUPT), China, in 2023. He is currently pursuing his M.E. degree at the same institution. His research interests include network security and privacy, with a specific focus on AI and blockchain technology.

Fu Xiao (Senior Member, IEEE) received the PhD degree in computer science and technology from the Nanjing University of Science and Technology, Nanjing, China, in 2007. He is currently a professor and a PhD supervisor with the School of Computers, Nanjing University of Posts and Telecommunications. His main research interests include wireless sensor networks and mobile computing. He is a member of the Association for Computing Machinery.

Yanchun Zhang (Member, IEEE) received the Ph.D. degree in computer science from the University of Queensland, Gatton, QLD, Australia, in 1991. He is currently a distinguished professor with Zhejiang Normal University, China and Emeritus Professor at Victoria University, Australia. He is Foreign Academician of the Russian Academy of Natural Sciences (RANS), and Fellow of Royal Society of Medicine of United kingdom (FRSM). He is the founding Director of Centre for Applied Informatics at Victoria University. His research interests include databases, data mining, social networking, web services and e-health / health informatics. He has published over 400 research papers in international journals and conference proceedings including ACM Transactionson Computer and Human Interaction (TOCHI), IEEE Transactions on Knowledge and Data Engineering (TKDE), VLDBJ and ICDE conferences as well as medical journals.